Beyond Implicit Bias: The Insignificance of SGD Noise in Online Learning

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Motivation: Online Learning Context

- Online learning (single epoch) dominates LLM training (e.g., GPT-2, Llama)
- Prior work assumes SGD noise provides implicit bias advantages universally
- Critical question: Does SGD noise matter in online learning?

Golden Path Hypothesis

- Proposed: SGD path $\approx$ noiseless GD path in online learning
- Contrast with "Fork in the Road" (offline: noise changes final model)
- Figure 2: Illustrates both scenarios with key implications

Offline vs Online Learning Contrast

- **Offline learning (multi-epoch):**
  - Smaller batch sizes → better minima (flatter, generalize better)
  - Figure 1 (left): Lower batch size plateaus at lower loss
- **Online learning (single epoch):**
  - Larger batch sizes → better performance (no implicit bias)
  - Figure 1 (right): Higher batch size reaches lower loss

Theoretical Support (Convex Case)

- **Equality of paths (Theorem 2.1):** In the online convex/quadratic setting, the expected SGD iterate equals the noiseless GD iterate at every step: E[w_SGD(t)] = w_GD(t). SGD noise only adds variance around the same trajectory.
- **Formalizes the golden path hypothesis:** In the convex case, noise introduces no new implicit bias in expectation.
- **Implication:** Batch size in online convex training affects optimization noise and compute — not the learned function's implicit bias.

The Non-Convex Case: SGD Noise Is Still Insignificant — Empirical Evidence

Observed, not proven: In realistic non-convex neural networks, the golden path hypothesis is observed empirically — on ResNet-18 (CIFAR-5m), ConvNeXt-T (ImageNet), and GPT-2-small (C4) — SGD noise is largely insignificant, and small batch sizes confer no implicit bias advantage in online learning.

- **Key conditions.** The empirical findings hold under natural, practical assumptions:
  - Online regime: data is processed in a single epoch, with no repeated passes
  - Natural training settings: standard deep architectures and typical optimization choices
  - Performance compared per gradient step, controlling for computational cost
  - Closeness measured in loss and function space, not weight space
- **Golden Path Hypothesis.** SGD takes noisy steps along the noiseless GD "golden path" — noise affects speed and stability, not the destination. This is a hypothesis supported by experiments, not a formal non-convex theorem.
- **Empirical validation.** The authors validate these claims on modern architectures and datasets across vision and language tasks — evidence rather than a proof.

Formal Statement: Online Setting & the Golden Path

Online learning setting

- **Single epoch:** each data point is seen exactly once; no repeated passes (fixed, non-repeating data ordering)
- **No generalization gap:** train and test distributions are identical — yet optimization remains non-convex
- **Per-step comparison:** performance measured per gradient step, controlling for computational cost

Main result: proven vs observed

Proven — Theorem 2.1 (convex case)

Quadratic loss $L(w) = w^T H w$ with $H \geq 0$; stochastic gradient $g(w) = \nabla L(w) + \xi$, $\xi \sim N(0, \sigma^2 I)$; fixed learning-rate schedule. Then $E[w_SGD(t)] = w_GD(t)$ at every step, and higher noise gives higher expected loss.

Closeness notion — Golden Path Hypothesis

(i) Loss trajectories: dropping noise at time t_o, the loss curve snaps to the low-noise curve within $\tau \ll t_o$ steps. (ii) Function space: average TV distance of softmax probabilities on the test set — after noise reduction, TV $\approx$ two independent low-noise runs, not $\approx$ high-noise vs low-noise.

Empirically observed (non-convex)

The golden path holds in realistic deep networks (ResNet-18/CIFAR-5m, ConvNext-T/ImageNet, GPT-2-small/C4) — evidence, not a theorem.

Snapping Back to Golden Path

- **Experiment:** Reduce noise at T_o steps (increase batch size)
- **Result (Figure 4):**
  - Loss sharply improves → follows low-noise trajectory
  - "Snaps" to golden path within $\tau \ll T_o$ steps
- Contrasts with offline learning (no snapping at convergence)

Function Space Evidence

- Measured total variation (TV) distance between models
- Golden path prediction:
  - After noise reduction, TV $\approx$ two low-noise models
  - Not $\approx$ high-noise vs low-noise models
- Figure 6: Confirms TV distance matches baseline

Implications & Conclusion

- **Practitioners:** Batch size choice driven by computation, not implicit bias
- **Theorists:** Focus noise effects on offline learning regime
- **Takeaway:** SGD in online learning = noisy GD steps on "golden path"